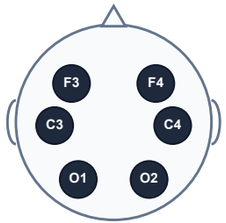
Stage 1: Bedside Acquisition & Masking

Sparse 6-Channel Telemetry

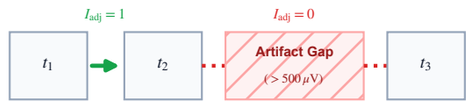


10-20 Sparse

CAR Reference

1,000 Hz / 4-s Epochs

Physical Adjacency Masking



$$I_{adj}(t_i, t_{i+1}) = \mathbf{1}_{\{t_{i+1} - t_i = 1\}}$$

Zero Cross-Gap Leakage

Clean Epochs & Masks

Stage 2: Dual Orthogonal Extraction

Temporal Sequence Memory

$$\rho_1 = \mathrm{Corr}(G_{t_i}, G_{t_{i+1}} \mid I_{adj} = 1)$$

$$D_1 = -\tfrac{1}{M}\sum |\Delta G| / (\bar{G} + \varepsilon)$$

Autoregressive Mesocircuit Continuity

Spatial Decoupling via OLS

$$\mathbf{P}^{\perp} = \mathbf{I} - \mathbf{X}_c(\mathbf{X}_c^{\top}\mathbf{X}_c)^{-1}\mathbf{X}_c^{\top}$$

$$r_{residual} = \tfrac{1}{3}\sum_{(L,R)} \mathrm{Corr}(\boldsymbol{\varepsilon}_L, \boldsymbol{\varepsilon}_R)$$

OLS Decoupling of Volume Conduction ($R^2 = 47\%$)

Orthogonal Features

Stage 3: Fault-Tolerant Fusion

4-Feature Orthogonal Suite

$$\mathbf{x} = [c_{global},\ -\mathrm{CV}_{\theta},\ \rho_1,\ r_{residual}]^{\top}$$

Spectral	Variability	Temporal	Spatial
c_{glob}	$-\mathrm{CV}_{\theta}$	ρ_1	r_{res}

Domain-Decoupled Orthogonal Suite

T-GECF Volatility Wrapper

$$\beta(z_v) = [1 + \exp(-\kappa(z_v - \theta_0))]^{-1}$$

$$S_{T-GECF} = \beta \tilde{S}_{rank} + (1 - \beta)\tilde{S}_{linear}$$

Low Volatility ($\beta \to 0$)	High Volatility ($\beta \to 1$)
Linear Consensus (0.9324)	Rank Consensus (Robust)

Unsupervised Collinearity Pruned ($|r| > 0.90$)

Fused Score S_{T-GECF}

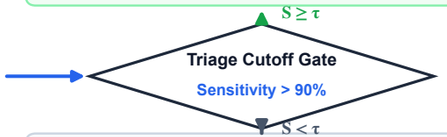
Stage 4: ICU Clinical Triage Funnel

Actionable Bedside Triage

Preserved Reverberation (CMD Candidate)

→ **Priority Confirmatory fMRI / ^{18}F-FDG PET**

Halves Secondary Transport Risks ($\approx 50\%$)



Depressed Dynamics

→ **Longitudinal Bedside EEG Monitoring**

Safeguards Against Premature WLST

Prospective Screening & Safe Triage Protocol